J1
12V INPUT
+12V_IN
GND
VIN
GND

Q1
AO4407A
+12V_PROTECTED
+12V_PROTECTED
+12V_PROTECTED
BP_GATE
+12V_IN
+12V_IN
+12V_IN
+12V_IN

R6
100k
BP_GATE
GND

D1
12V gate clamp
+12V_PROTECTED
BP_GATE

U1
TPS54536J1802FR
+12V_PROTECTED
BOOT1
BOOT2
SW_BUS
SW_BUS
GND1
FB1
SW_BUS
SW_BUS
GND1
EN1
SW_BUS
SW_BUS
SW_BUS
SW_BUS
+12V_PROTECTED
+12V_PROTECTED
GND1
GND1
GND1
GND1
SYN1
ABOOT
CBOOT
SW
VLDOIN
VCC
AGND
FB
VOUT1
VOUT2
AGND
RT
PG
SPSP
SYNC_MODE
NC
SYN
SYN2
PGND
PGND
AGND
AGND

PWR_FLAG
+12V_PROTECTED

PWR_FLAG
GND

C1
10uF 50V X7R
+12V_PROTECTED
GND

C2
10uF 50V X7R
+12V_PROTECTED
GND

C3
47uF 10V X7R
SW_BUS
GND

C9
47uF 10V X7R
SW_BUS
GND

C10
47uF 10V X7R
SW_BUS
GND

R1
24.9k 0.1%
FB1
GND

R7
102k 0.1%
SW_BUS
FB1

R8
15.8k 1%
RT
GND

R9
100k
SW_BUS
PG

C11
100nF 25V
SW_BUS
GND

C12
47uF 25V bulk
+12V_PROTECTED
GND

U7
TPS25570RB
GND
SW_BUS
SW_BUS
SLEW1
VBUS1
VBUS1
FAULT1
GND
GND
IN
IN
EN
ILIM
OUT
OUT
FAULT
EP

R2
32.4k 0.1%
SLEW1
GND

R4
100k
SW_BUS
FAULT1

C4
100nF 25V
SW_BUS
GND

J2
USB-A 3A
VBUS1
DN1
DP1
GND1
GND1
VBUS
D-
D+
GND
SHIELD

U5
USBLC6-2SC6
DN1
GND
DP1
DP1
VBUS1
DN1
IO1
GND
IO2
IO2
VBUS
IO1

U3
TPS25570RB
GND
SW_BUS
SW_BUS
SW_BUS
ILIM2
VBUS2
VBUS2
FAULT2
GND
GND
IN
IN
EN
ILIM
OUT
OUT
FAULT
EP

R3
32.4k 0.1%
ILIM2
GND

R5
100k
SW_BUS
FAULT2

C5
100nF 25V
SW_BUS
GND

C7
22uF 10V
VBUS2
GND

J3
USB-A 3A
VBUS2
DN2
DP2
GND2
GND2
VBUS
D-
D+
GND
SHIELD

U6
USBLC6-2SC6
DN2
GND
DP2
DP2
VBUS2
DN2
IO1
GND
IO2
IO2
VBUS
IO1

C6
22uF 10V
VBUS1
GND

U4
TPS2513A3DVR
DP1
DN1
DP2
DN2
SW_BUS
GND
DP1
GND
DP2
DN2
IN
DM1

C8
100nF 25V
SW_BUS
GND

C13
47uF 10V X7R
SW_BUS
GND

C14
47uF 10V X7R
SW_BUS
GND

H1
M3 MOUNT

H2
M3 MOUNT

H3
M3 MOUNT

H4
M3 MOUNT

Sheet: /
File: USB_POWER.kicad_sch
Title: UMI circuit development
Size: A1 | Date:
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Rev: D6 — THREE BOARD PRO
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